

## RESEARCH ARTICLE

# An Efficient Parallel Branch Network for Multi-Class Classification of Prostate Cancer From Histopathological Images

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## ABSTRACT

Prostate cancer is one of the prevalent forms of cancer, posing a significant health concern for men. Accurate detection and classification of prostate cancer are crucial for effective diagnosis and treatment planning. Histopathological images play a pivotal role in identifying prostate cancer by enabling pathologists to identify cellular abnormalities and tumor characteristics. With the rapid advancements in deep learning, Convolutional Neural Networks (CNNs) have emerged as a powerful tool for tackling complex computer vision tasks, including object detection, classification, and segmentation. This paper proposes a Parallel Branch Network (PBN), a CNN architecture specifically designed for the automatic classification of prostate cancer into its subtypes from histopathological images. The paper introduces a novel Efficient Residual (ER) block that enhances feature representation using residual learning and multi-scale feature extraction. By utilizing multiple branches with different filter reduction ratios and dense attention mechanisms, the block captures diverse features while preserving essential information. The proposed PBN model achieved a classification accuracy of 93.16% on the Prostate Gleason dataset, outperforming all other comparison models.

## 1 | Introduction

Prostate cancer is the second most common cancer in men, with about 1 in 8 men being diagnosed. As of 2021, approximately 3.4 million men in the U.S. were living with prostate cancer Siegel et al. [1]. In 2024, it is projected that 299 010 new cases of prostate cancer are diagnosed in the U.S., accounting for 14.9% of all new cancer cases Siegel et al. [2] Quayle [3]. Early detection and accurate diagnosis of prostate cancer are crucial for improving patients recovery rate as the aggressiveness of the disease increases with its grade. The grade of prostate cancer is determined by how abnormal the cells appear. Higher-grade cancers are more aggressive and likely to spread more quickly, whereas lower-grade cancers tend to grow more slowly. Therefore,

identifying the cancer grade early allows doctors to make treatment planning more effective, leading to better survival rates for patients. Diagnostic methods such as computed tomography (CT) Buzug [4] Li et al. [5], magnetic resonance imaging (MRI) Shukla-Dave and Hricak [6], Prostate-Specific Antigen (PSA) testing Pezaro et al. [7], and Digital Rectal Exam (DRE) Schröder et al. [8] are commonly used traditional techniques to detect and evaluate prostate cancer. Other methods, including Transrectal Ultrasound (TRUS) Aarnink et al. [9], prostate biopsy Matlaga et al. [10], and advanced imaging techniques like Positron Emission Tomography (PET) Schöder and Larson [11] and Bone Scintigraphy Suh et al. [12], are also employed to assess the spread of cancer. The primary limitation of these techniques is the lack of sensitivity and specificity, especially

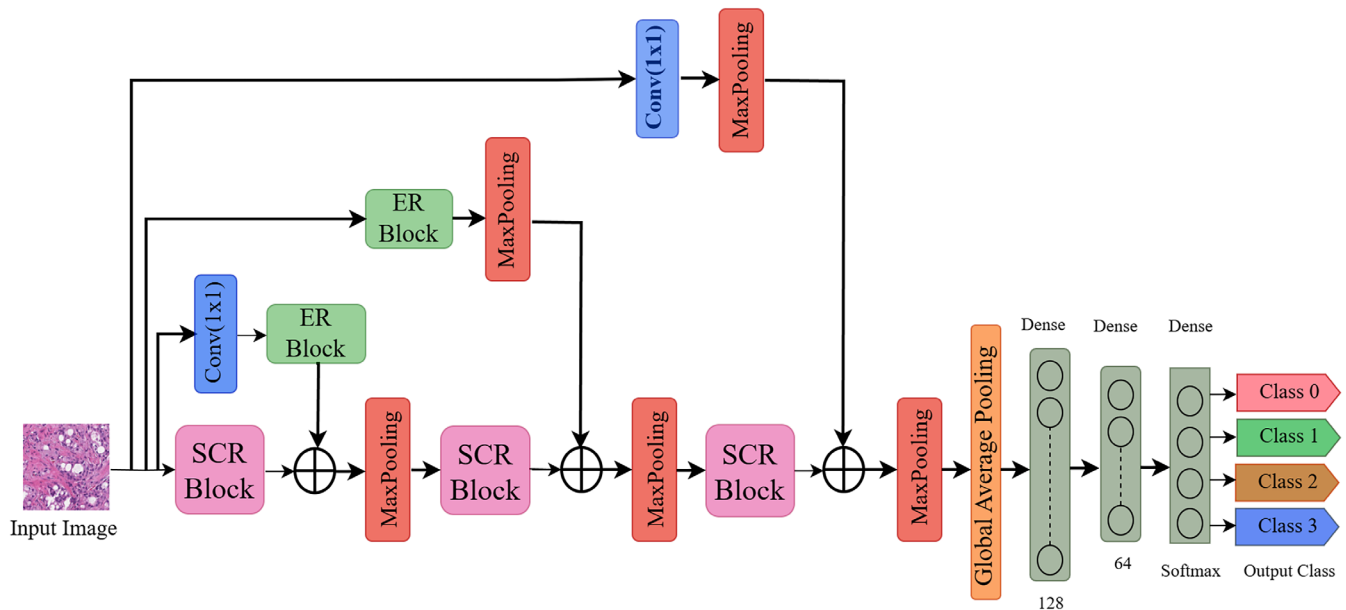
in detecting early-stage or low-grade tumors. Additionally, these methods heavily rely on expert interpretation, which is subjective and can vary between different observers. To overcome these limitations, histopathological analysis Matoso and Epstein [13] is considered a reliable method for grading prostate cancer. Pathologists examine the tissue samples under the microscope to identify the cellular abnormalities and assign gleason scores Gleason and Mellinger [14]. However, manual grading is time-consuming and labor-intensive, highlighting the need for automated and objective diagnostic solutions. The Gleason grading system is a widely used method for assessing the severity of prostate cancer by examining the cancer cells under a microscope. It involves assigning two grades: the primary grade (representing the most common cell pattern) and the secondary grade (representing the second most common pattern), each scored from 1 to 5. The total Gleason score is the sum of these two grades, with a range from 6 to 10, where higher scores indicate more aggressive cancer. A score of 6 (3 + 3) indicates low-risk or less aggressive cancer, while a score of 7 (3 + 4 or 4 + 3) suggests moderate risk. Scores between 8 and 10 reflect high-risk cancer, indicating a more aggressive form of the disease. These scores are then categorized into four grades: Grade 0 for Gleason scores less than 6, Grade 1 for a score of 6 (low risk), Grade 2 for a score of 7 (moderate risk), and Grade 3 for scores between 8 and 10 (high risk).

The structure of the rest of the paper is as follows. Related studies are presented in Section 2. The design of the proposed PBN model is discussed in Section 3. The training and implementation details are discussed in Section 4. The simulation results and discussions are covered in Section 5. The conclusion of this paper is presented in Section 6.

## 2 | Related Studies

Several studies have explored early detection methods for prostate cancer, starting with traditional techniques such as Prostate-Specific Antigen (PSA) testing Chen et al. [15]. These studies show that larger prostates have lower detection rates due to the higher presence of low-volume cancers. This often causes PSA levels to rise from benign glandular elements rather than malignancies. The need for more sensitive and selective screening strategies was further emphasized by Schroder [16], in cases with low PSA levels, combining digital rectal examination and transrectal ultrasonography. Computer-based Gleason grading methods primarily involve extracting tissue features and employing feature-engineered classifiers to distinguish between benign and cancerous tissues. A similarity measurement method for classifying histopathological slides of prostate cancer was performed using Fourier transforms and the Karhunen-Loève expansion technique Chien and Fu [17]. A limitation of this approach is that it heavily relies on the availability of an initial set of preclassified images, which requires the expertise of trained professionals. A novel Statistical Representation of Homology Profile (SRHP) Yan et al. [18] approach to automate Gleason grading of prostate cancer was given that achieved high performance in distinguishing Gleason Grades 3 and 4. In recent years, deep convolutional neural networks (DCNNs) have been increasingly utilized for prostate cancer detection. Wang et al. [19] showcased the ability of DCNNs to identify prostate

cancer, highlighting their performance in comparison to traditional techniques such as SIFT and Bag of Words (BoW). An automated CNN-based framework was designed to detect clinically significant prostate cancer using axial Diffusion-Weighted Imaging (DWI) Yoo et al. [20]. The advent of deep learning (DL) has significantly changed the medical imaging field, providing powerful tools for automating complex tasks such as image classification and segmentation. Convolutional Neural Networks (CNNs), in particular, have demonstrated exceptional success in learning hierarchical patterns from medical images, making them highly effective for cancer detection and grading. Models such as VGG16 Simonyan and Zisserman [21] and ResNet He et al. [22] have demonstrated remarkable capabilities in image classification tasks, with ResNet addressing challenges like vanishing gradients through innovative residual layers and skip connections. These skip connections allow information to bypass certain layers, ensuring efficient gradient flow in deep neural networks. Iqbal et al. [23] compared deep learning models like ResNet-101 He et al. [22] and LSTM Staudemeyer and Morris [24] with traditional machine learning models such as SVM Hearst et al. [25], KNN Laaksonen and Oja [26], and Decision Trees Quinlan [27], concluding that ResNet-101 outperformed the non-deep learning methods in detecting prostate cancer. Based on this, the ResHist model Gour et al. [28], a 152-layer residual convolutional network, has been specifically designed to classify breast cancer histopathological images into benign and malignant categories by extracting rich and discriminative features. The InceptionNet Szegedy et al. [29] architecture introduced the concept of multi-scale feature extraction by applying convolutional operations with filters of varying sizes within the same layer. Building on the base Inception architecture, Xception further enhanced its depth by introducing six intermediate layers, creating an “extreme inception” design. This additional depth allows for better hierarchical feature learning Chollet [30]. However, all these models are computationally expensive, requiring significant memory and processing power. DenseNet Huang et al. [31] revolutionized connectivity by enabling feature maps from all preceding layers to be reused in subsequent layers, promoting better gradient flow and feature reuse across the network. The MFR-DenseNet architecture extends DenseNet by incorporating multiple feature reweighting mechanisms. It integrates squeeze-and-excitation blocks at various layers to adaptively reweight channel-specific feature responses. The model combines Channel Feature Reweight DenseNet (CFR-DenseNet) and Inter-Layer Feature Reweight DenseNet (ILFR-DenseNet) using ensemble learning, enhancing performance across various medical imaging tasks. While MFR-DenseNet improves accuracy, the added complexity of integrating multiple mechanisms increases training time and makes hyperparameter tuning more challenging Zhang et al. [32]. Recent advancements in deep learning have introduced attention mechanisms to enhance the capabilities of CNNs. By integrating techniques like the Channel Block Attention Mechanism (CBAM), models like Livernet Chanchal et al. [33] Aatresh et al. [34] and BreastNet Toğaçar et al. [35] focus on the most important features in medical images, effectively reducing redundant information and enhancing feature representation. Both of them have a high number of parameters, which makes them a complex model. Shufflenet Zhang et al. [36] introduced the concept of channel shuffling, which involves dividing channels and shuffling them to promote information diversity.



**FIGURE 1** | Architecture of the proposed PBN model.

However, it has a huge number of parameters and FLOPs, making it a heavy and complex model. This concept was later incorporated into the RCCGNet model that employs the Shared Channel Residual (SCR) block Chanchal et al. [37] for the classification of kidney cancer Lal et al. [38]. This has the concept of shuffling between higher and lower layers. It has a lesser number of parameters compared to all other models. TurkerNet, a lightweight CNN model for skin cancer classification, employs a squeeze-and-excitation mechanism along with a combination of efficient decomposition and residual blocks to improve feature extraction Tuncer et al. [39]. It has the least number of FLOPs compared to benchmark models but is not so effective in accurately predicting the classes. The Enhanced Lightweight CNN (EL-CNN) integrates spatial pyramid pooling and CBAM modules for colorectal cancer (CRC) detection across various datasets Pan et al. [40]. Transfer learning methods, using pre-trained models like Xception, DenseNet121, and VGG16, have also been employed to improve model performance on histopathological datasets Kumar et al. [41] Tan et al. [42]. The integration of CNNs with Transformers is also effective. The FocalNeXt model, a hybrid architecture that combines the feature extraction capabilities of ConvNeXt with the attention mechanisms of FocalNet, has produced good results in automated lung cancer detection Gulsoy and Kablan [43]. Another hybrid architecture that involves Enhanced Convolutional Learning Modules (ECLMs), a multi-level Attention Learning Module (ALM), and Transitional Modules (TMs) forms a comparable framework for classifying hematoxylin and eosin-stained colon cancer images Dabass et al. [44]. Another contribution to prostate cancer grading is the Prostate Grading Network (ProsGradNet), which integrates a Context Guided Shared Channel Residual (CGSCR) block for feature extraction by using channel splitting and clustering techniques Prabhu et al. [45]. Keeping these architectures in mind, a novel deep CNN model has been developed to further enhance classification accuracy while maintaining a low computational complexity.

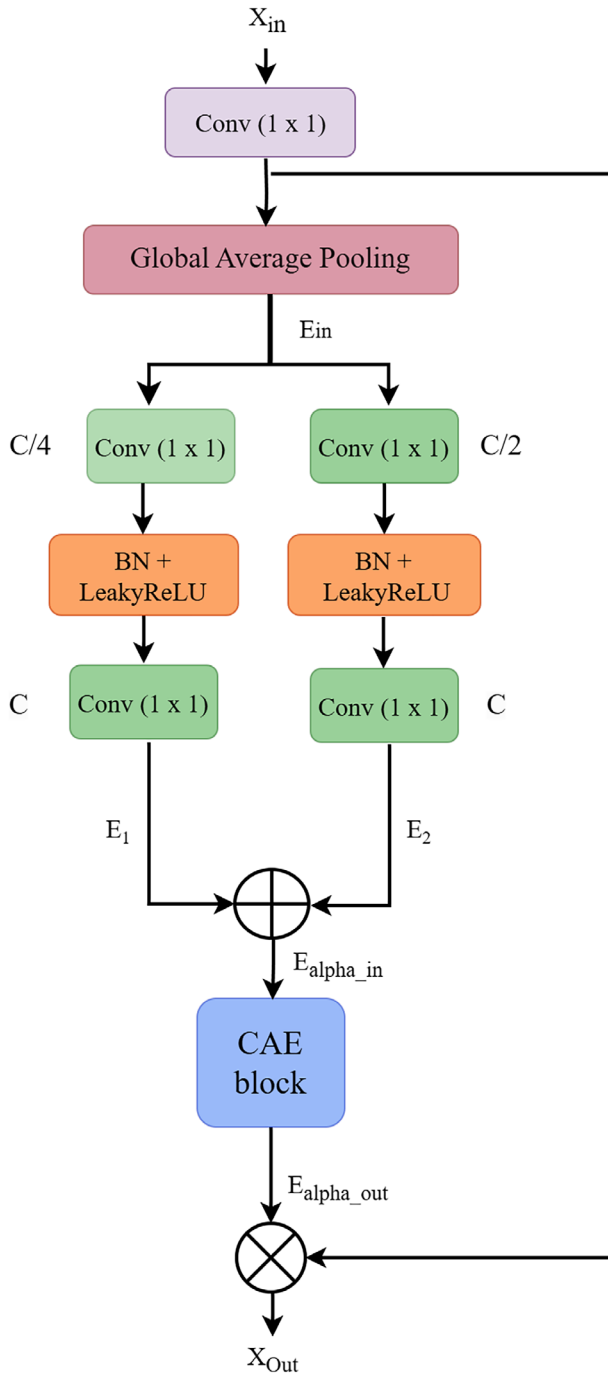
The primary contributions of the paper are:

- Introduced an Efficient Residual (ER) block within the architecture that significantly boosts the overall performance using residual learning and feature enhancement techniques.
- Designed a novel deep learning based CNN model to improve the classification accuracy of prostate cancer grading.

### 3 | Proposed Model

This section discusses the detailed design aspect of the proposed Parallel Branch Network (PBN) model. The proposed PBN model, as illustrated in the Figure 1, takes an input of size  $256 \times 256 \times 3$  and is processed through multiple parallel branches. Initially, the input is passed through a Shared Channel Residual (SCR) block Chanchal et al. [37], which employs information sharing between channels of both lower and higher layer feature maps. Simultaneously, the input undergoes  $(1 \times 1)$  convolution followed by an ER block as shown in Figure 2. The outputs from these branches are then added and downsampled using maxpooling layer.

In the next stage, the architecture processes the output again through parallel branches. One branch consists of an SCR block, whereas the other branch has an ER block followed by a maxpooling layer. The results of these branches are added and further Maxpooled. In the final stage of feature extraction, the output is processed through two additional parallel branches. One branch has the SCR block and the second branch has a  $(1 \times 1)$  convolution followed by the maxpooling layer. The outputs from these branches are also added, and the result is processed through a GlobalAveragePooling layer to reduce spatial dimensions into a compact 1D vector. This vector is then passed through two fully connected layers: the first with 128 units and the second with 64 units, each followed by ReLU and a LeakyReLU layer respectively to add non-linearity. Finally, the architecture ends with a softmax



**FIGURE 2** | Efficient residual (ER) block.

layer that outputs probabilities across four classes. The use of parallel branches with ER and SCR blocks, combined with maxpooling and addition, ensures robust feature extraction and efficient learning.

### 3.1 | Efficient Residual Block

The block diagram of the novel Efficient Residual (ER) block is depicted in Figure 2. This novel block improves feature representation using residual learning that helps to preserve previous information. The input tensor  $X_{in}$  initially undergoes  $(1 \times 1)$  convolution and is passed through a Global Average Pooling layer to

reduce the spatial dimensions of the tensor into a single feature vector  $E_{in}$  per channel as given in Equation (1), where GAP is Global Average Pooling and  $\text{Conv}_{(1 \times 1)}$  is convolution operation of kernel size  $(1 \times 1)$ .

$$E_{in} = \text{GAP}(\text{Conv}_{(1 \times 1)}(X_{in})) \quad (1)$$

The reshaped tensor is then passed through two separate branches. Branch 1 applies a  $(1 \times 1)$  convolution with number of filters reducing to one fourth followed by batch normalization, a leaky ReLU activation, and another  $(1 \times 1)$  convolutional layer to yield  $E_1$ . Branch 2 follows a similar sequence but reduces the number of filters into half instead of one-fourth to yield  $E_2$ . The mathematical representations of  $E_1$  and  $E_2$  are given in Equations (2) and (3), where  $\sigma$ ,  $C$  and BN represents sigmoid activation function, number of channels or filters and Batch Normalization respectively.

$$E_1 = \sigma(\text{Conv}_{(1 \times 1)}(\text{LeakyReLU}(\text{BN}(\text{Conv}_{(1 \times 1)}(E_{in}, C/4))), C) \quad (2)$$

$$E_2 = \sigma(\text{Conv}_{(1 \times 1)}(\text{LeakyReLU}(\text{BN}(\text{Conv}_{(1 \times 1)}(E_{in}, C/2))), C) \quad (3)$$

Having two branches with different reduction ratios allows the model to learn diverse set of features. The output of both branches are merged using addition operation to give  $E_{\alpha in}$  as in Equation (4), which is then passed to the Channel-wise Alpha Efficient (CAE) block, which extracts multi scale features through parallel transformations and modulating channels to produce  $E_{\alpha out}$ . This enables the network to enhance relevant features while suppressing less significant ones, improving the overall representation of features.

$$E_{\alpha in} = E_1 + E_2 \quad (4)$$

### 3.2 | Channel-Wise Alpha Efficient Block

The block diagram of the Channel-wise Alpha Efficient (CAE) block is shown as in Figure 3. This block applies a Global Average Pooling to the fused output  $E_{\alpha in}$  to give  $EA_{in}$  as in Equation (5).

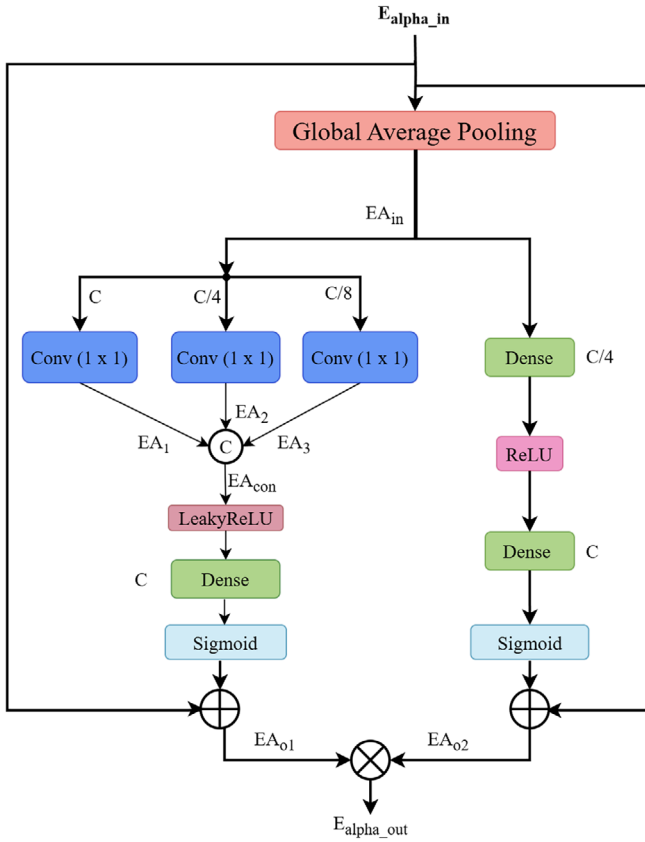
$$EA_{in} = \text{GAP}(E_{\alpha in}) \quad (5)$$

Again, the reshaped tensor is fed into two parallel branches. First branch has three parallel paths. Path 1 retains the full filter count, preserving the original channel information that gives  $EA_{in1}$ . Path 2 reduces the filters to one-fourth, focusing on abstract features to give  $EA_{in2}$ , whereas path 3 reduces the filters to one-eighth, capturing finer details that gives  $EA_{in3}$ . The mathematical equations representing  $EA_{in1}$ ,  $EA_{in2}$ , and  $EA_{in3}$  are presented in Equations (6–8) respectively.

$$EA_{in1} = \text{Conv}_{(1 \times 1)}(EA_{in}, C) \quad (6)$$

$$EA_{in2} = \text{Conv}_{(1 \times 1)}(EA_{in}, C/4) \quad (7)$$

$$EA_{in3} = \text{Conv}_{(1 \times 1)}(EA_{in}, C/8) \quad (8)$$



**FIGURE 3** | Channel-wise alpha efficient (CAE) block.

The outputs of these three paths are then concatenated to produce  $EA_{con}$  as in Equation (9), where leakyReLU is applied to introduce nonlinearity to model complex relationships effectively, where  $\odot$  denotes concatenation operator.

$$EA_{conc} = EA_{in1} \odot EA_{in2} \odot EA_{in3} \quad (9)$$

Further, the concatenated output is given to a dense layer with sigmoid activation function that generates attention weights for each channel by assigning scores according to their importance. Channels with higher attention weights are then amplified, while those with lower weights are suppressed. Another feature refinement is carried out in the second branch of the same block using a dense attention mechanism followed by a residual addition. The first dense layer reduces the number of filters into  $(C/r)$ , where  $r$  equals to 4, which helps the model to learn a lower-dimensional representation of the reshaped tensor. The second dense layer expands the filter size back to  $C$ . The output of the second dense layer is then added to the original input tensor using the addition operator. This residual addition ensures that the original features are not lost during the feature extraction process, while also incorporating the learned modifications from the dense layers. The outputs from both branches  $EA_{o1}$  and  $EA_{o2}$  are then multiplied to get the  $E_{alpha\_out}$ . The equations representing  $EA_{o1}$  and  $EA_{o2}$  are given in Equation (10) and Equation (11).

$$EA_{o1} = \sigma(\text{Dense}(\text{LeakyRelu}(EA_{conc}), C)) + E_{alpha\_in} \quad (10)$$

$$EA_{o2} = \sigma(\text{Dense}(\text{ReLU}(\text{Dense}(EA_{in}, C/4)), C)) + E_{alpha\_in} \quad (11)$$

## 4 | Training and Implementation

This section describes the training and implementation procedures of the proposed PBN model that covers the dataset, hyperparameters used, and performance evaluation criteria. The grading pipeline of the proposed PBN model and other comparison models is illustrated in Figure 4.

### 4.1 | Prostate Gleason Dataset

The Prostate Gleason Dataset includes prostate cancer biopsy images taken from the PANDA dataset Bulten et al. [46]. It contains around 70000 image patches, each sized  $256 \times 256$  pixels. The samples are classified into four categories based on their Gleason score or grade as given in Table 1.

The original distribution of the dataset is presented in Table 2. Due to resource limitation, it is difficult to train all samples of this original dataset. So, a custom dataset was further created whose distribution is provided in Table 3. A total of 11684 samples were selected for training, 2854 for testing, and 2939 for validation. The visualization of all four classes of the dataset is shown in Figure 5.

### 4.2 | Training Setup

A workstation with 16GB GPU and 64GB CPU with Intel XEON processor is used for training of the comparison model as well as proposed PBN model. All the models are trained for 100 epochs, with a batch size of 4. The optimizer used is Adam optimizer, with a learning rate scheduler implemented to adjust the learning rate dynamically based on a patience value of eight epochs. Early stopping is employed with a patience of 30 epochs to prevent overfitting. The categorical cross entropy (CCE) loss function is used for optimizing the model's performance Chanchal et al. [37]. The mathematical equation for CCE is given as in Equation (12), where  $C$  denotes the number of classes,  $y_i$  are the true label for  $i^{th}$  class and  $f(x)_i$  represents the predicted probability for class  $i$ , given the input  $x$ .

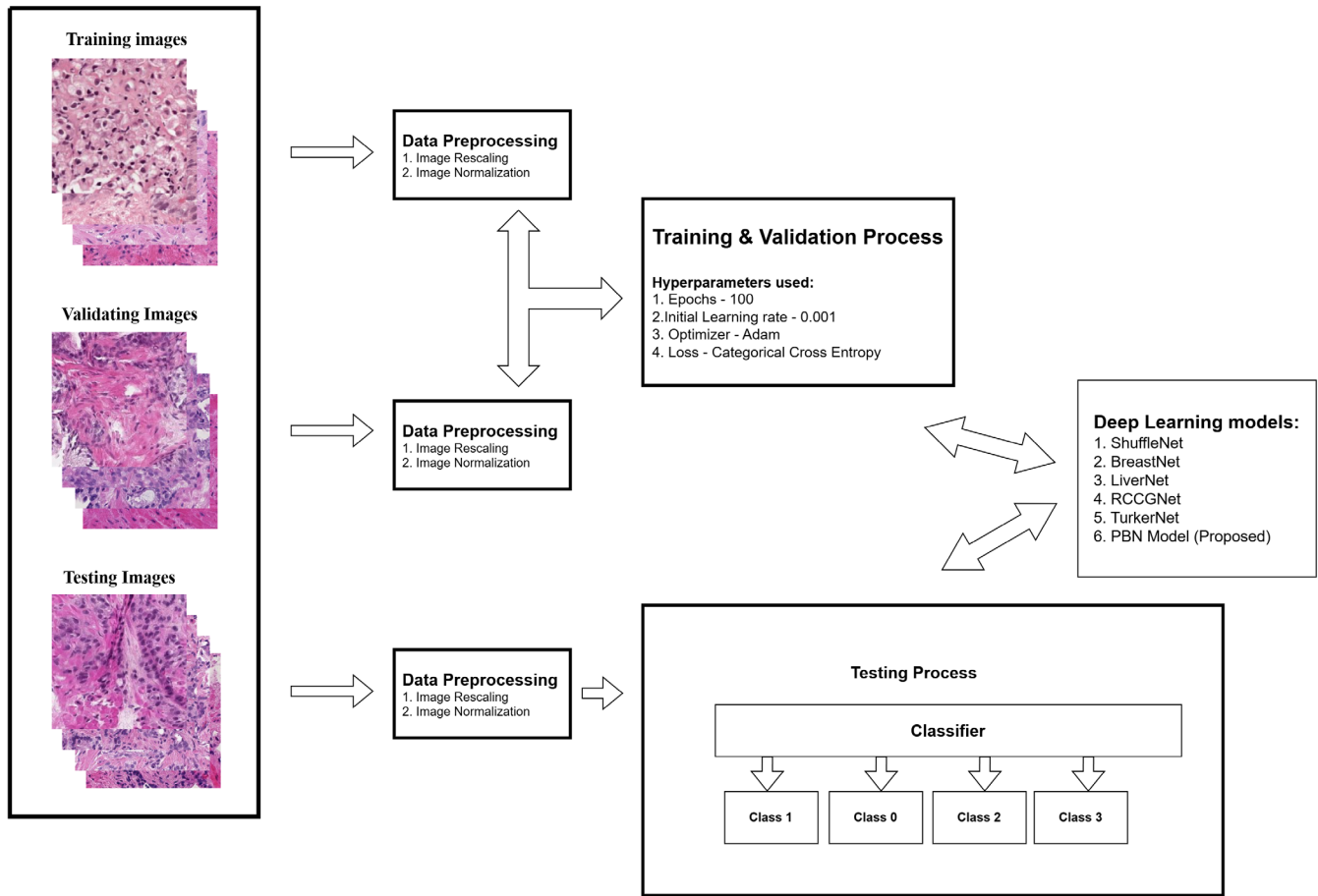
$$CCE = - \sum_{i=1}^C y_i \log(f(x)_i) \quad (12)$$

## 5 | Results and Discussion

This section discusses ablation study, comparative studies, computational complexity analysis followed by statistical analysis of the proposed PBN model. It also highlights the significant improvement in the performance of the proposed PBN model.

### 5.1 | Ablation Study

The ablation study is conducted to evaluate the contribution of individual components in the proposed PBN model. Four intermediate models were trained, and the results are given in Table 4.



**FIGURE 4** | Grading pipeline of proposed PBN model.

**TABLE 1** | Different classes and their gleason scores.

| Classes | Grades  | Gleason score           |
|---------|---------|-------------------------|
| Class 0 | Grade 1 | Gleason score $\leq 6$  |
| Class 1 | Grade 2 | Gleason score 7 (4 + 3) |
|         | Grade 3 | Gleason score 7 (3 + 4) |
| Class 2 | Grade 4 | Gleason score 8         |
| Class 3 | Grade 5 | Gleason score 9–10      |

**TABLE 2** | Distribution of the original Prostate Gleason dataset.

| Classes      | Training samples | Test samples  | Validation samples |
|--------------|------------------|---------------|--------------------|
| Class 0      | 11 200           | 2800          | 2240               |
| Class 1      | 11 200           | 2800          | 2240               |
| Class 2      | 7936             | 1985          | 1590               |
| Class 3      | 10814            | 2704          | 2163               |
| <b>Total</b> | <b>41 150</b>    | <b>10 289</b> | <b>8233</b>        |

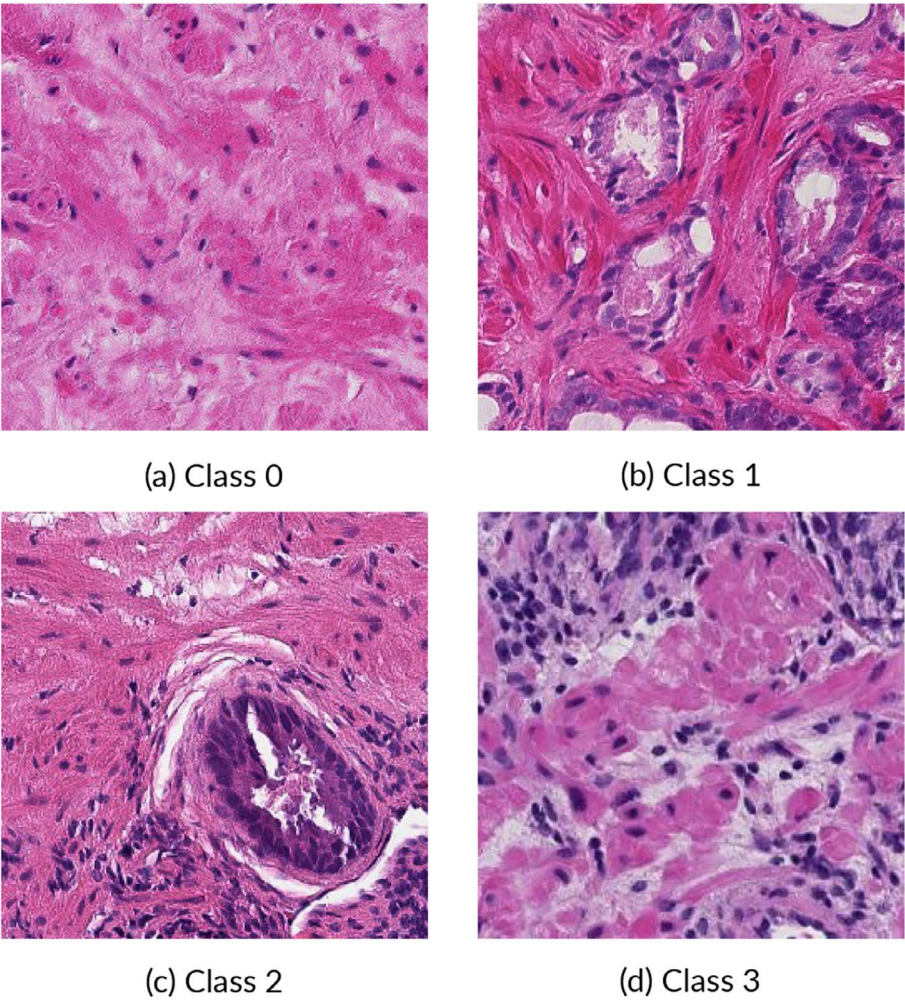
Intermediate model 1: It involves the removal of the ER block from the proposed PBN model. It gives an accuracy of 92.43% while precision, recall, and F1-Score are 92.51%, 92.49%, and 92.49% respectively. In the absence of the ER block, the model

**TABLE 3** | Distribution of the custom Prostate Gleason dataset.

| Classes      | Training samples | Test samples | Validation samples |
|--------------|------------------|--------------|--------------------|
| Class 0      | 3018             | 754          | 800                |
| Class 1      | 3000             | 700          | 728                |
| Class 2      | 2851             | 700          | 718                |
| Class 3      | 2815             | 700          | 693                |
| <b>Total</b> | <b>11 684</b>    | <b>2854</b>  | <b>2939</b>        |

retains only the SCR block Chanchal et al. [37] along with two dense layers. When comparing the proposed PBN model with Intermediate Model 1, a decrease in performance is observed, with accuracy dropping by 0.73%, precision by 0.77%, recall by 0.69%, and F1 score by 0.73%. Hence, the removal of the ER block from the architecture resulted in a decrease in performance.

Intermediate model 2: In this model, the CAE Block is removed from the proposed PBN architecture. The model attains an accuracy of 92.39%, with precision, recall, and F1-score values of 92.50%, 92.43%, and 92.46%, respectively. Compared to intermediate model 2, there is an improvement of 0.77% in accuracy, 0.78% in precision, 0.75% in recall, and 0.76% in F1-score in the proposed PBN model. This indicates the effect of the CAE block in the architecture, which, when removed, decreased the performance.



**FIGURE 5** | Visualization of four different classes of prostate cancer.

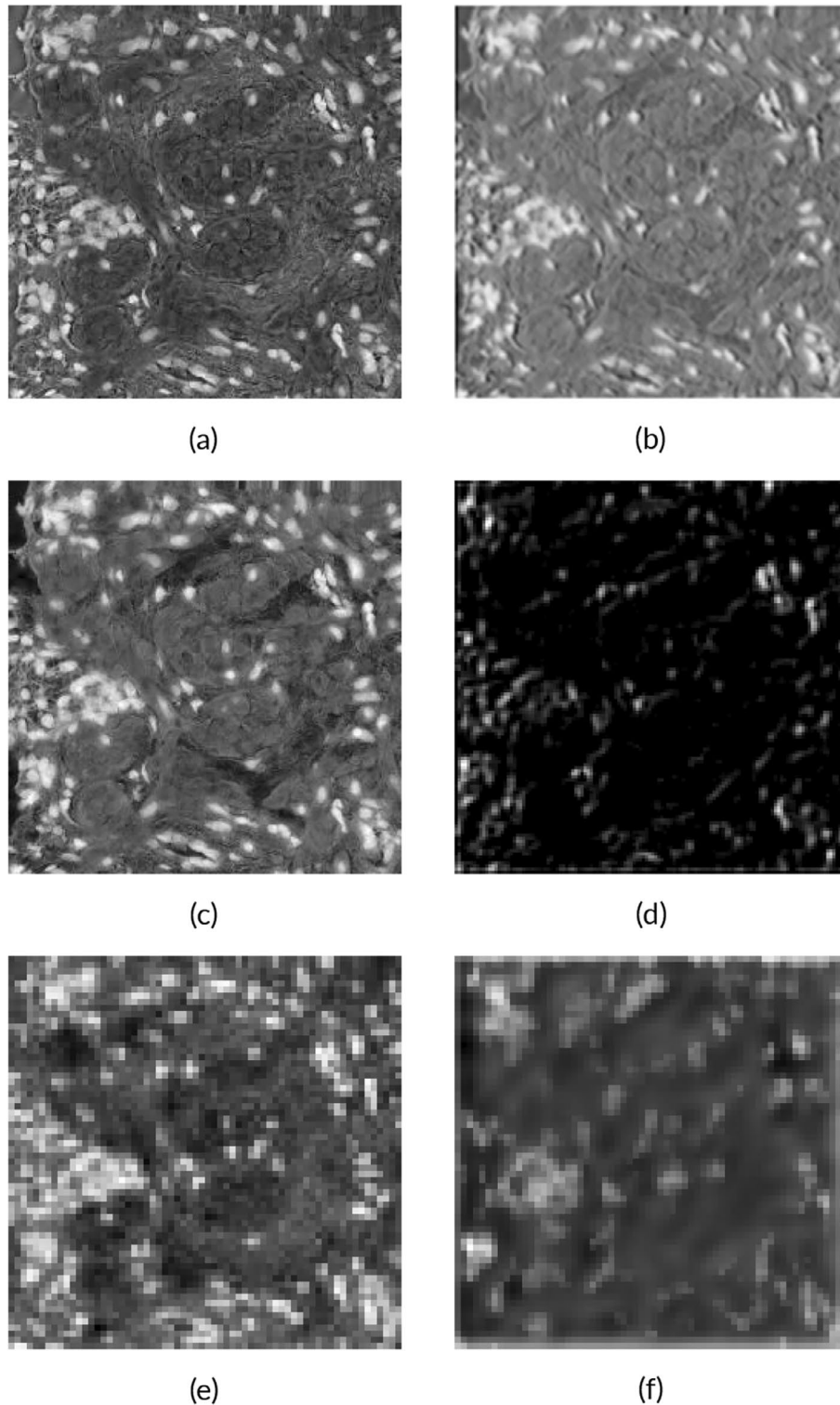
**TABLE 4** | Ablation study of proposed PBN model on custom Prostate Gleason Dataset.

| Metrics   | Intermediate model 1 | Intermediate model 2 | Intermediate model 3 | Proposed PBN model |
|-----------|----------------------|----------------------|----------------------|--------------------|
| Precision | 0.9251               | 0.9250               | 0.8633               | <b>0.9328</b>      |
| Recall    | 0.9249               | 0.9243               | 0.8611               | <b>0.9318</b>      |
| F1-score  | 0.9249               | 0.9246               | 0.8619               | <b>0.9322</b>      |
| Accuracy  | 0.9243               | 0.9239               | 0.8605               | <b>0.9316</b>      |

Intermediate model 3: It involves the removal of the SCR block from the architecture. Intermediate Model 3 achieves a lower performance across all metrics, with an accuracy of 86.05%, precision of 86.33%, recall of 86.11%, and F1-score of 86.19%. Compared to the proposed PBN model, the accuracy, precision, recall, and F1-score of the intermediate model 3 have reduced by 7.11%, 6.95%, 7.07%, and 7.03%. This performance gap highlights the significant role of the SCR block in enhancing feature representation. The absence of channel shuffling reduces the model's ability to extract and combine diverse feature information, leading to a less robust overall performance.

The feature maps of the proposed PBN model are visualized at six different intermediate stages as shown in

Figure 6. These feature maps reveal how the model gradually learns from simple features to complex patterns that enable accurate predictions. Structural changes like glandular distortion and irregularly shaped nuclei indicates poorly differentiated cancer cells which are associated with a high gleason score. Figure 6a,c represent the feature maps obtained after the first and second ER blocks, respectively. Figure 6b,d,f represent the feature maps obtained after the first, second, and third SCR block, respectively. From the figures, it is evident that the nuclei get more focused and highlighted after each deeper layer. At deeper layers, the nuclei become irregular and enlarged. Also, we can observe that higher scores are assigned to the nuclei in stages 3 and 5.



**FIGURE 6** | Visualization of extracted features at different layers of proposed model.

## 5.2 | Comparison With Benchmark Models

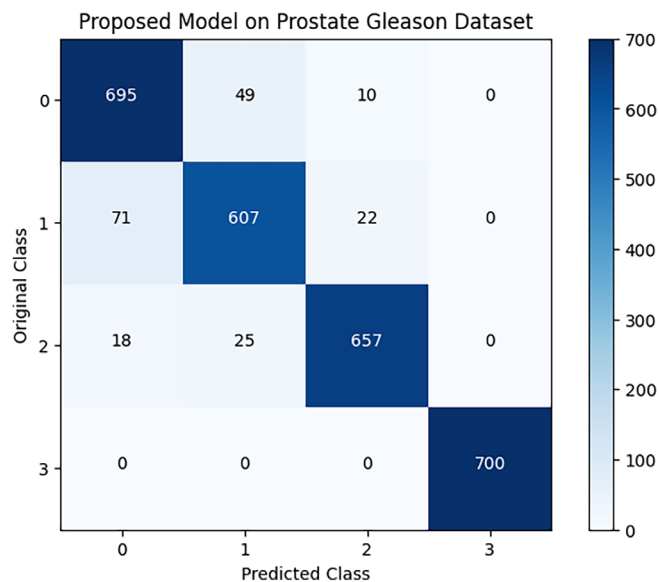
The performance evaluation of the proposed PBN model against five existing deep learning models ShuffleNet, BreastNet, LiverNet, RCCGNet, and TurkerNet on the Prostate Gleason Dataset is presented in Table 5. The results demonstrate that the proposed PBN model consistently outperforms the other models across all evaluation metrics, including precision, recall, F1-score, and overall accuracy.

Accuracy is one of the pivotal metrics in evaluating the classification performance of any deep learning models. The proposed PBN model achieves an overall accuracy of 93.16%, outperforming RCCGNet (91.86%), ShuffleNet (92.01%), TurkerNet (90.50%), LiverNet (90.32%) and BreastNet (86.72%). This improvement highlights the effectiveness of the proposed architecture in accurately classifying the Prostate Gleason Dataset across all classes. For precision, the proposed PBN model achieves an overall score of 93.28%, surpassing

**TABLE 5** | Performance evaluation of proposed PBN model and other existing DL models on custom Prostate Gleason Dataset.

| Metrics   | Class   | ShuffleNet (2018) | BreastNet (2020) | LiverNet (2021) | RCCGNet (2023) | TurkerNet (2024) | Proposed PBN Model |
|-----------|---------|-------------------|------------------|-----------------|----------------|------------------|--------------------|
| Precision | 0       | 0.8641            | 0.8344           | 0.8560          | 0.8758         | <b>0.9046</b>    | 0.8864             |
|           | 1       | 0.8782            | 0.7427           | 0.8649          | 0.8629         | 0.7909           | <b>0.8913</b>      |
|           | 2       | 0.9455            | 0.9008           | 0.8953          | 0.9456         | 0.9402           | <b>0.9535</b>      |
|           | 3       | 1.0               | 0.9985           | 1.0             | 1.0            | 0.9971           | <b>1.0</b>         |
|           | Overall | 0.9219            | 0.8691           | 0.9040          | 0.9211         | 0.9082           | <b>0.9328</b>      |
| Recall    | 0       | <b>0.9363</b>     | 0.8355           | 0.9151          | 0.9350         | 0.8554           | 0.9217             |
|           | 1       | 0.8242            | 0.7671           | 0.7685          | 0.8214         | <b>0.87</b>      | 0.8671             |
|           | 2       | <b>0.9785</b>     | 0.8700           | 0.9285          | 0.9206         | 0.8985           | 0.9385             |
|           | 3       | 1.0               | 0.9985           | 1.0             | 1.0            | 1.0              | <b>1.0</b>         |
|           | Overall | 0.9197            | 0.8678           | 0.9030          | 0.9192         | 0.9060           | <b>0.9318</b>      |
| F1-score  | 0       | 0.8987            | 0.8349           | 0.8846          | <b>0.9044</b>  | 0.8793           | 0.9037             |
|           | 1       | 0.8504            | 0.7547           | 0.8139          | 0.8416         | 0.8285           | <b>0.8790</b>      |
|           | 2       | 0.9318            | 0.8851           | 0.9116          | 0.9329         | 0.9189           | <b>0.9460</b>      |
|           | 3       | 1.0               | 0.9985           | 1.0             | 1.0            | 0.9985           | <b>1.0</b>         |
|           | Overall | 0.9202            | 0.8683           | 0.9025          | 0.9197         | 0.9063           | <b>0.9322</b>      |
| Accuracy  | 0       | 0.9442            | 0.9127           | 0.9369          | 0.9462         | 0.9379           | <b>0.9481</b>      |
|           | 1       | 0.9288            | 0.8777           | 0.9138          | 0.9234         | 0.9117           | <b>0.9414</b>      |
|           | 2       | 0.9670            | 0.9446           | 0.9558          | 0.9676         | 0.9611           | <b>0.9737</b>      |
|           | 3       | 1.0               | 0.9992           | 1.0             | 1.0            | 0.9992           | <b>1.0</b>         |
|           | Overall | 0.9201            | 0.8672           | 0.9032          | 0.9186         | 0.9050           | <b>0.9316</b>      |

ShuffleNet (92.19%), RCCGNet (92.11%) and TurkerNet (90.82%). This can be attributed to the integration of the ER block with enhanced channel shuffling from the SCR block, which provides more effective feature refinement. The recall values further explain the superior performance of the proposed PBN model, with an overall score of 0.9318, which is again higher than ShuffleNet (91.97%), RCCGNet (91.92%) and LiverNet (90.30%). When considering the F1-score, which balances precision and recall, the proposed PBN model again leads with an overall score of 93.22%, showing an improvement over the other models. It can also be observed that the second-best performing model is RCCGNet, which achieves competitive performance across all metrics. TurkerNet achieves the highest precision for Class 0 of 90.46% and for Class 1 of 79.09%. In contrast, RCCGNet excels in recall for class 0 with a value of 93.50% and attains the best F1 score for class 0 of 90.44%. Although these models yield good results for certain metrics in specific classes, the proposed PBN model consistently achieves superior overall performance across all key metrics. The integration of the ER block with a shared channel shuffling mechanism in the proposed PBN model contributes significantly to its overall performance. This combination allows for more effective feature extraction and concurrent parameter learning. The ER and SCR blocks work synergistically to improve feature extraction. The SCR block promotes

**FIGURE 7** | Confusion Matrix of proposed PBN model.

information sharing between channels, allowing the model to learn more diverse and informative features. The ER block, using residual learning, helps in preserving prior information

while also refining the feature representation. The CAE block further enhances the model's performance by focusing on multi-scale feature extraction. It applies parallel transformations to modulate and refine features, allowing the network to focus on the most relevant information while suppressing less important ones. Together, these three blocks allow the model to perform better in terms of accuracy, precision, and other key metrics, distinguishing it from other models and contributing to its outstanding results.

The provided confusion matrix in Figure 7 shows the classification performance of the proposed PBN model across four distinct classes. For Class 0, the proposed PBN model accurately classified 695 samples as belonging to this category, with only 59 samples misclassified into other classes. In the

case of Class 1, the proposed PBN model classified 607 samples correctly, and 93 samples were misclassified. For Class 2, the proposed PBN model correctly classified 657 samples, with only 43 misclassifications. Similarly, Class 3, which represents high-risk cancer cases, showed perfect classification. All 700 samples were accurately identified without any false positives or false negatives.

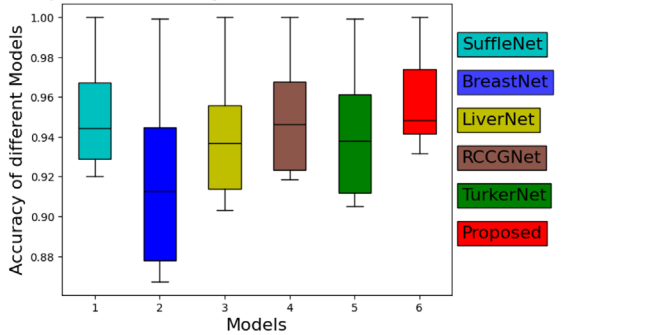
### 5.3 | Computational Complexity Analysis

The computational complexity of the proposed PBN model and comparison models is presented in Table 6, providing a detailed evaluation of parameters, FLOPs (Floating Point Operations) and testing time. The proposed PBN model has the smallest number of parameters among all the models, with just 0.361 M parameters. This demonstrates its compact architecture, which reduces memory requirements without compromising performance. In comparison, ShuffleNet has the highest parameter count of 23.718 M, making it significantly more resource-intensive. Both the proposed PBN model and RCCGNet have 5.88 G FLOPs, indicating comparable computational demands during training and inference. TurkerNet achieves the lowest FLOPs of 0.59 G, whereas ShuffleNet has the highest FLOPs of 8.69 G, reflecting its complex architecture. While the proposed PBN model achieves state-of-the-art results, its test time is 30 milliseconds, slightly higher than RCCGNet (22 ms) and TurkerNet (14 ms). This increase can be attributed to the inclusion of the ER block, which enhances feature extraction. One of the limitations of the model thus is the higher test time compared to other comparison models, which needs to be addressed in the future.

**TABLE 6** | Computational complexity of Models.

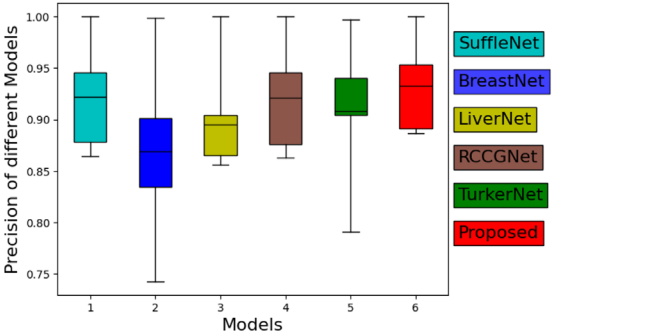
| Models                    | Parameters (M) | FLOPs (G)   | Test time (ms) |
|---------------------------|----------------|-------------|----------------|
| ShuffleNet                | 23.718         | 8.69        | 20             |
| BreastNet                 | 0.603          | 5.66        | 16             |
| LiverNet                  | 0.598          | 4.97        | 29             |
| RCCGNet                   | 0.363          | 5.88        | 22             |
| TurkerNet                 | 1.584          | 0.59        | 14             |
| <b>Proposed PBN model</b> | <b>0.361</b>   | <b>5.88</b> | <b>30</b>      |

Comparison of Accuracy of different models on Prostate Gleason dataset



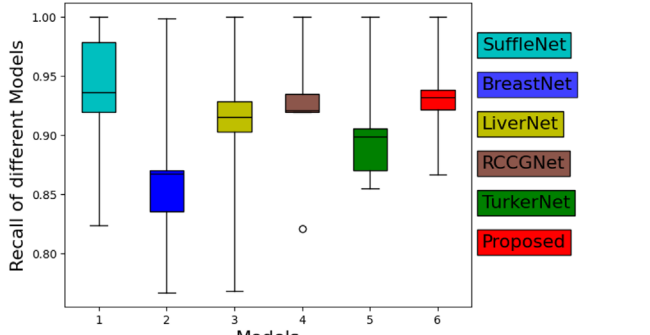
(a) Accuracy Box Plot

Comparison of Precision of different models on Prostate Gleason dataset



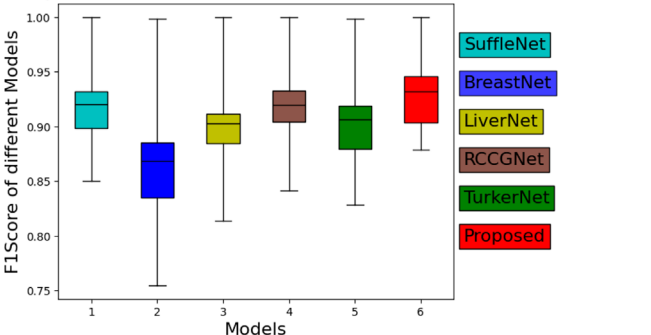
(b) Precision Box Plot

Comparison of Recall of different models on Prostate Gleason dataset



(c) Recall Box Plot

Comparison of F1-Score of different models on Prostate Gleason dataset



(d) F1 Score Box Plot

**FIGURE 8** | Box Plots of Performance Metrics: (a) Accuracy, (b) precision, (c) recall, and (d) F1 Score.

## 5.4 | Statistical Analysis Using Box Plot

A boxplot is a graphical tool that displays the spread and central tendency of data, highlighting the median and interquartile range (IQR). It effectively identifies variability, skewness, and outliers, making it ideal for comparing data distributions. The boxplot for each of the four performance metrics on the Prostate Gleason dataset is presented in Figure 8. The median values of accuracy, precision, F1-score, and recall of the proposed PBN model are higher than those of the other comparison models, indicating superior and consistent performance. Also, the proposed PBN model exhibits a narrower interquartile range (IQR) for accuracy and recall, indicating more consistent and stable performance.

The proposed PBN model demonstrates superior advantages in terms of accuracy and computational efficiency. It achieves the highest accuracy among all the existing DL models without much increase in complexity. This high performance is attributed to the combination of ER, SCR, and CAE blocks that enhance feature representation by focusing on multiscale features. Compared to other existing DL models, the proposed PBN model has the lowest number of parameters (0.361 M). In terms of FLOPs, it has the third least number of FLOPs, after TurkerNet and BreastNet. However, as a limitation, the proposed PBN has a test time of 30 ms, which is slightly higher than other comparison models. Also, the generalizability of the model can be improved by testing the model on other datasets as well.

## 6 | Conclusion

This study presented a novel deep learning-based CNN model, PBN, for prostate cancer grading from histopathology images, focused on enhancing feature extraction and representation through the use of three building blocks: the Shared Channel Residual (SCR) block, the Efficient Residual (ER) Block, and the Channel-wise Alpha Efficient (CAE) block. These components, when integrated into the network architecture, enabled effective multi-scale feature extraction, leading to superior performance. The proposed PBN model gave an accuracy of 93.16% on the custom Prostate Gleason dataset, outperforming existing benchmark models. Despite having the lowest parameter count, the test time of the proposed PBN model is higher compared to other models. Further optimization of the proposed PBN model could be done to reduce the test time without compromising accuracy.

### Author Contributions

**Vishal Srivastava:** writing – original draft, conceptualization, methodology, investigation. **Akshaya Prabhu:** conceptualization, writing – review and editing. **Sravya Nedungatt:** conceptualization, writing – review and editing. **K. Vibha Damodara:** conceptualization, writing – review and editing. **Shyam Lal:** conceptualization, writing – review and editing, supervision. **Jyoti Kini:** resources, data curation, formal analysis, validation, writing – review and editing.

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### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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